

PUV Pipeline — Methods and Validation Results

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1 Methods

1.1 Nearshore wave measurements

Nearshore wave and current dynamics were measured using bottom-mounted Nortek Vector acoustic Doppler velocimeters (ADV) configured in pressure–velocity (PUV) mode, sampling continuously at 2 Hz. Instruments were deployed along MOP (Monitoring and Prediction) transect lines at nominal depths of 5–7 m, with two cross-shore positions per transect. Deployment metadata—including instrument heading, clock drift, and pressure sensor height above the bed (z_s , typically 0.75–0.77 m)—were obtained from field deployment notes.

1.2 Level 1: Quality control and coordinate rotation

Raw burst files (.dat/.sen/.hdr) were merged into continuous time series at the 2 Hz sampling rate. Clock drift corrections were applied linearly over each deployment based on measured drift at recovery. A battery cutoff detection algorithm identified intermittent sampling caused by battery depletion by flagging gaps exceeding 1 s between consecutive sensor records; all data following the first such gap were discarded.

Quality control filters removed samples where: (1) pitch or roll exceeded $\pm 5^\circ$; (2) *pressure fell outside the range [2], where is the deployment median pressure*; or (3) *the minimum beam correlation dropped below 70%, following Nortek*

A variability-based tilt quality control was applied rather than a fixed angular threshold. The rolling standard deviation of pitch and roll was computed over 60-second windows; samples where the variability exceeded 2° were flagged as unstable. An absolute tilt cap of 30° was retained for severely compromised instruments. Samples with stable but non-zero tilt (e.g., from pipes bent during storm events) were retained and corrected via a sample-by-sample three-dimensional rotation using the measured pitch (ϕ) and roll (θ) angles:

$$\mathbf{v}_{\text{corrected}} = \mathbf{R}_{\text{roll}}(\theta) \mathbf{R}_{\text{pitch}}(\phi) \mathbf{v}_{\text{raw}}, \quad (1)$$

where

$$\mathbf{R}_{\text{roll}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}, \quad \mathbf{R}_{\text{pitch}} = \begin{bmatrix} \cos \phi & 0 & \sin \phi \\ 0 & 1 & 0 \\ -\sin \phi & 0 & \cos \phi \end{bmatrix}.$$

This tilt correction was applied before the heading rotation to buoy coordinates, ensuring that the velocity vector was first returned to the local vertical frame before projection onto the geographic coordinate system.

Velocity components were rotated from the instrument XYZ frame to a geographic buoy coordinate system ($+x$ west, $+y$ north, $+z$ up; left-handed, consistent with the MOP convention) using the magnetic heading from deployment notes and the magnetic declination computed via the International Geomagnetic Reference Field (IGRF-13) model at the deployment location and epoch.

1.3 Level 2: Spectral analysis

1.3.1 Segmentation and preprocessing

The continuous L1 time series were divided into non-overlapping 17.1 min segments of 2048 samples at 2 Hz. Segments in which more than 10% of pressure or velocity samples were flagged as NaN were discarded; for segments with $\leq 10\%$ NaN, gaps were filled by linear interpolation before further processing. Prior to spectral analysis, velocities were rotated to a shore-normal coordinate

frame (+ x onshore, + y alongshore north) using the shore-normal angle retrieved from the CDIP THREDDS server for the corresponding MOP station.

For each valid segment, the mean pressure was used to compute the total water column depth $H = \bar{P} \times 10^4 / (\rho g) + z_s$, where $\bar{P}$ is the segment-mean pressure in $\rho = 1025 \text{ kg/m}^3$, $g = 9.81 \text{ m/s}^2$, and z_s is the pressure sensor height above the bed. Mean cross-shore, alongshore, and vertical velocities and temperature were recorded for each segment prior to detrending. The pressure time series was converted from meters of water head, and all channels (pressure, u , v , w) were linearly detrended.

1.3.2 Spectral estimation

Power spectral densities were estimated using Welch’s method [welch1967] with 256-sample sub-segments (128 s), a Hanning window, and 50% overlap, yielding approximately 34 effective degrees of freedom per segment. This configuration provides a frequency resolution of $\Delta f \approx 0.0078 \text{ Hz}$ over the range 0–1 Hz. Cross-spectra between pressure and the two horizontal velocity components were computed using the same windowing parameters for directional analysis.

1.3.3 Pressure correction

The measured pressure spectrum was converted to a surface elevation spectrum using the linear wave theory transfer function:

$$K_p(f) = \frac{\cosh(kz_s)}{\cosh(kH)} \quad (2)$$

where $k(f)$ is the wavenumber obtained by solving the linear dispersion relation $\omega^2 = gk \tanh(kH)$ via Newton’s method. The surface elevation spectrum was computed as $S_\eta(f) = S_p(f)/K_p^2(f)$. Frequency bins where $K_p < 0.1$ were set to zero to suppress noise amplification at high frequencies; the resulting cutoff frequency f_{cut} was recorded for each segment.

1.3.4 Bulk wave parameters

Significant wave height was computed as $H_s = 4\sqrt{m_0}$, where $m_0 = \int S_\eta(f) df$ is the zeroth spectral moment, integrated separately over the sea–swell band (0.04–0.25 Hz) and infragravity band (0.004–0.04 Hz). Peak period $T_p = 1/f_p$ was determined from the frequency of maximum S_η within the sea–swell band, and the mean zero-crossing period was computed as $T_{m02} = \sqrt{m_0/m_2}$. The energy-weighted mean wave direction was obtained from the first-harmonic directional coefficients a_1 and b_1 derived from the pressure–velocity cross-spectra. Total energy flux was computed as $F = \rho g \int c_g(f) S_\eta(f) df$, where c_g is the group velocity from linear wave theory.

1.3.5 Near-bed velocity and stress

Near-bed orbital velocities were estimated by scaling the measured velocity spectrum at sensor height to the bed using linear wave theory. Each frequency bin of the velocity FFT was multiplied by $\cosh(kH)/\cosh[k(H-z_s)]$, and the result was inverse-transformed to obtain the near-bed velocity time series [[e.g.,] [guza1980]. The RMS near-bed orbital velocity U_b and wave orbital excursion A_w were computed from the bed velocity time series for each segment.

Wave-induced bed shear stress was estimated using the Swart (1974) friction factor formulation with a bed roughness of $k_s = 2.5D_{50}$, where $D_{50} = 0.25 \text{ mm}$ (to be updated with site-specific grain size distributions from laser particle analysis). Reynolds stress components ($\overline{u'w'}$, $\overline{v'w'}$, $\overline{u'v'}$) and turbulent kinetic energy were computed from the detrended velocity fluctuations. Velocity

moment statistics including skewness S_k and acceleration asymmetry A_s were computed following elgar1985,hoefel2003 for characterizing wave nonlinearity relevant to sediment transport.

1.4 Validation against MOP wave model

PUV-derived wave parameters were validated against hourly wave data from the CDIP Monitoring and Prediction (MOP) system, a spectral wave model initialized by offshore CDIP buoy observations. The MOP model provides wave energy density spectra $S_\eta(f)$ at 10 m depth along each MOP transect line. MOP spectra were shoaled to the PUV instrument depth using linear wave theory energy conservation:

$$S_\eta(f, h_{\text{PUV}}) = S_\eta(f, 10 \text{ m}) \times \frac{c_g(f, 10 \text{ m})}{c_g(f, h_{\text{PUV}})} \quad (3)$$

Shoaled MOP wave heights were computed by integrating the shoaled spectrum over the sea-swell band using the (non-uniform) MOP frequency bandwidth vector. PUV segment values (17 min resolution) were interpolated to the MOP hourly time grid for comparison. Agreement was quantified using R^2 , RMSE, bias (PUV – MOP), and the Willmott (1981) skill score.

Frequency-dependent validation was performed by computing the mean spectral ratio $S_\eta^{\text{PUV}}(f)/S_\eta^{\text{MOP,shoaled}}(f)$ across all matched hours, binned into low-swell (0.04–0.08 Hz), mid-swell (0.08–0.14 Hz), and high-swell (0.14–0.25 Hz) bands. This diagnostic isolates whether systematic biases originate from the pressure transfer function (which amplifies high-frequency energy most aggressively) or from other sources.

To investigate the origin of systematic biases, four candidate mechanisms were tested: (1) non-linear shoaling via triad interactions, assessed by correlating the spectral ratio with the Ursell number $Ur = H_s L_p^2 / h^3$; (2) directional narrowing during shoaling, assessed by comparing the first-harmonic directional spread $\sigma_1 = \sqrt{2(1 - \sqrt{a_1^2 + b_1^2})}$ between PUV and MOP; (3) bound long wave contamination, tested by examining the scaling of low-frequency excess energy with H_s^2 and the frequency structure of the excess relative to the expected group frequency; and (4) MOP model spectral peak broadening, quantified by comparing the Goda peakedness parameter $Q_p = (2/m_0^2) \int f S_\eta^2(f) df$ and the half-power spectral bandwidth between PUV and shoaled MOP spectra.

2 Results

2.1 PUV–MOP validation

2.1.1 Bulk wave parameter agreement

Of 40 instruments across 20 deployments, 33 were processed successfully (82.5%), with 7 failures attributable to hardware issues (burial, bent pipes, dead batteries). PUV-derived wave parameters showed strong agreement with the MOP wave model across all four TBR23 instruments (Table ??). For instruments unaffected by biofouling, H_s correlations exceeded $R^2 = 0.83$ with $\text{RMSE} < 0.06$ m and biases of $+0.002$ – $+0.008$ m. One instrument (MOP580 5m) exhibited lower agreement ($R^2 = 0.66$, bias = $+0.037$ m), attributed to intermittent kelp fouling that reduced the valid segment rate to 49% compared with 85–91 for the remaining instruments. A small but consistent positive H_s bias was observed across all instruments, with PUV-measured wave heights systematically exceeding shoaled MOP values.

2.1.2 Frequency-dependent spectral bias

The frequency-resolved spectral ratio $S_\eta^{\text{PUV}}(f)/S_\eta^{\text{MOP,shoaled}}(f)$, averaged across all matched hours, revealed that the positive H_s bias was concentrated at the swell spectral peak (0.04–0.08 Hz), where the pressure transfer function $K_p > 0.9$ and the correction is minimal (Table ??). The mid-swell band (0.08–0.14 Hz) showed near-unity ratios (0.99–1.03), and the high-swell band (0.14–0.25 Hz), where the pressure correction is most aggressive (K_p as low as 0.24), also showed only modest excess (1.03–1.05). The low-swell amplification increased with decreasing depth, from a band-averaged ratio of 1.17 at 8.3 m to 1.28 at 5.5 m. The overall excess energy relative to the shoaled MOP spectrum was 4.5 at 8.3 m depth and 8.7 at 5.5 m.

The concentration of excess energy at the spectral peak—where the pressure correction factor is close to unity—indicates that the bias does not originate from the pressure-to-surface-elevation transfer function.

2.1.3 Hypothesis testing

Four candidate mechanisms for the systematic positive bias were evaluated (Table ??).

Nonlinear shoaling. The Ursell number $Ur = H_s L_p^2 / h^3$, which characterizes wave nonlinearity and governs the strength of triad interactions during shoaling, showed no correlation with the low-swell spectral ratio ($R^2 = 0.004$, $N = 5228$ segment-hours across all instruments). The median Ursell number ranged from 12 at 8.3 m depth to 30 at 5.5 m, spanning conditions from weakly to moderately nonlinear.

Directional narrowing. The first-harmonic directional spread σ_1 , computed from the PUV cross-spectral a_1 and b_1 coefficients, was systematically *wider* than the MOP model spread at swell frequencies (15 ° vs. 10 ° at the peak). This is opposite to the narrowing expected from refraction-driven focusing. No correlation was found between the spread ratio and H_s amplification ($R^2 < 0.001$). The wider PUV spread may reflect scattered wave energy at shallow depths not represented in the forward-propagating MOP model.

Bound long waves. Bound wave energy forced by wave group modulation scales as H_s^2 [longuetig-gins1962]. The low-swell excess energy showed no positive correlation with H_s^2 ($R = -0.14$), and

the amplification ratio weakly *decreased* with increasing wave height ($R = -0.19$). Furthermore, the excess energy was concentrated at the swell spectral peak (0.04–0.08 Hz), not at the expected group/bound wave frequencies (~ 0.01 – 0.03 Hz).

MOP spectral peak broadening. The Goda peakedness parameter Q_p was systematically higher for PUV spectra than for shoaled MOP spectra (median $Q_p = 1.88$ vs. 1.75 , ratio 1.07). The PUV half-power spectral bandwidth was 16.7% narrower than MOP (0.039 Hz vs. 0.047 Hz), and the PUV peak spectral density was 14.4% higher. The peakedness difference $\Delta Q_p = Q_p^{\text{PUV}} - Q_p^{\text{MOP}}$ showed the strongest correlation with H_s amplification of any tested mechanism ($R = 0.44$, $R^2 = 0.20$). Hours when the PUV spectrum was more peaked relative to MOP corresponded to hours with larger positive H_s bias.

These results indicate that the MOP spectral wave model introduces numerical broadening of the spectral peak during propagation from the initialization depth (10 m) to the instrument depth. The in-situ PUV measurement captures the true, narrower spectral peak shape, resulting in higher peak energy density and thus higher H_s for a given total variance. This spectral broadening is a recognized property of discretized spectral wave models [e.g.,] rogers2002, booi1999 and has not been previously quantified using co-located PUV observations in this depth range.

2.1.4 Cross-deployment confirmation

The spectral peak broadening pattern was confirmed across 11 instruments spanning multiple deployments, sites, and seasons. The correlation between ΔQ_p and H_s amplification ranged from $R = 0.30$ to $R = 0.63$ across individual instruments, with all showing the same sign. These instruments spanned depths from 5.5 m to 15.5 m and included both summer and winter wave climates, indicating that the MOP model spectral broadening is a robust and general feature rather than a site-specific or seasonally dependent artifact.